Analysis of Covariance Tutorials: One-way and Factorial Analyses

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```
# Load in Required Libraries
library(psych)
library(tidyverse)
library(jmv)
library(ggpubr)
library(ggrain)
library(apaTables)
library(ez)
library(rstatix)
```

ANCOVA | One-Way Research Context Prompt

Research Context:

Researchers investigated the effect of exercises in reducing anxiety levels. Anxiety scores were measured from three groups of individuals practicing physical exercises at different levels (`low` , `moderate` , and `high`).

The anxiety score was measured `pre-` and 6-months `post-exercise` training programs. It is expected that any reduction in the anxiety by the exercises programs would also depend on the participant's baseline level of anxiety.

In this analysis we use the `pre-test anxiety score` as the covariate and are interested in possible differences between `group` with respect to the `post-test anxiety score` .

One-way ANCOVA | Data

```
dat_anx <- read.csv("ancova1.csv")
```

ANCOVA | Descriptive Stats (Baseline-test)

```
dat_anx$groupF <- factor(dat_anx$group,
                         levels = c("Low", "Moderate", "High"),
                         labels = c("Low", "Moderate", "High"))

# Descriptive Stats
describeBy(dat_anx$pretest,
           dat_anx$groupF,
           mat=TRUE)
```

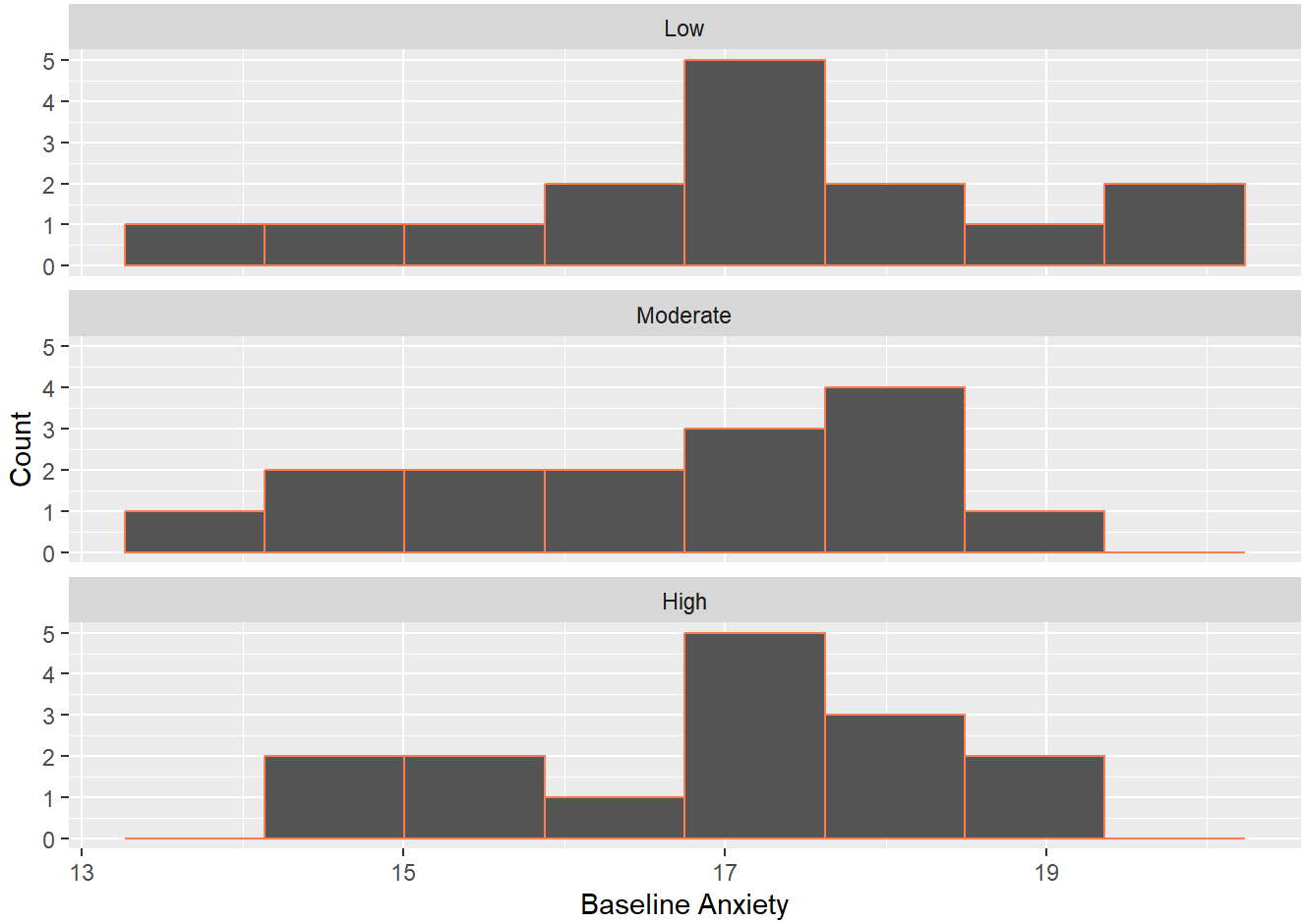
```
##     item   group1 vars  n     mean       sd median  trimmed     mad  min  max
## X11    1      Low    1 15 17.08667 1.628701   17.0 17.10769 1.33434 14.1 19.8
## X12    2 Moderate    1 15 16.64667 1.565643   16.9 16.66923 1.63086 13.7 19.3
## X13    3     High    1 15 17.01333 1.321183   17.3 17.04615 1.33434 14.6 19.0
##     range       skew   kurtosis        se
## X11   5.7 -0.1029298 -0.8254734 0.4205288
## X12   5.6 -0.2002025 -1.0718963 0.4042473
## X13   4.4 -0.3718028 -1.1608155 0.3411279
```

```
describeBy(dat_anx$posttest,
           dat_anx$groupF,
           mat=TRUE)
```

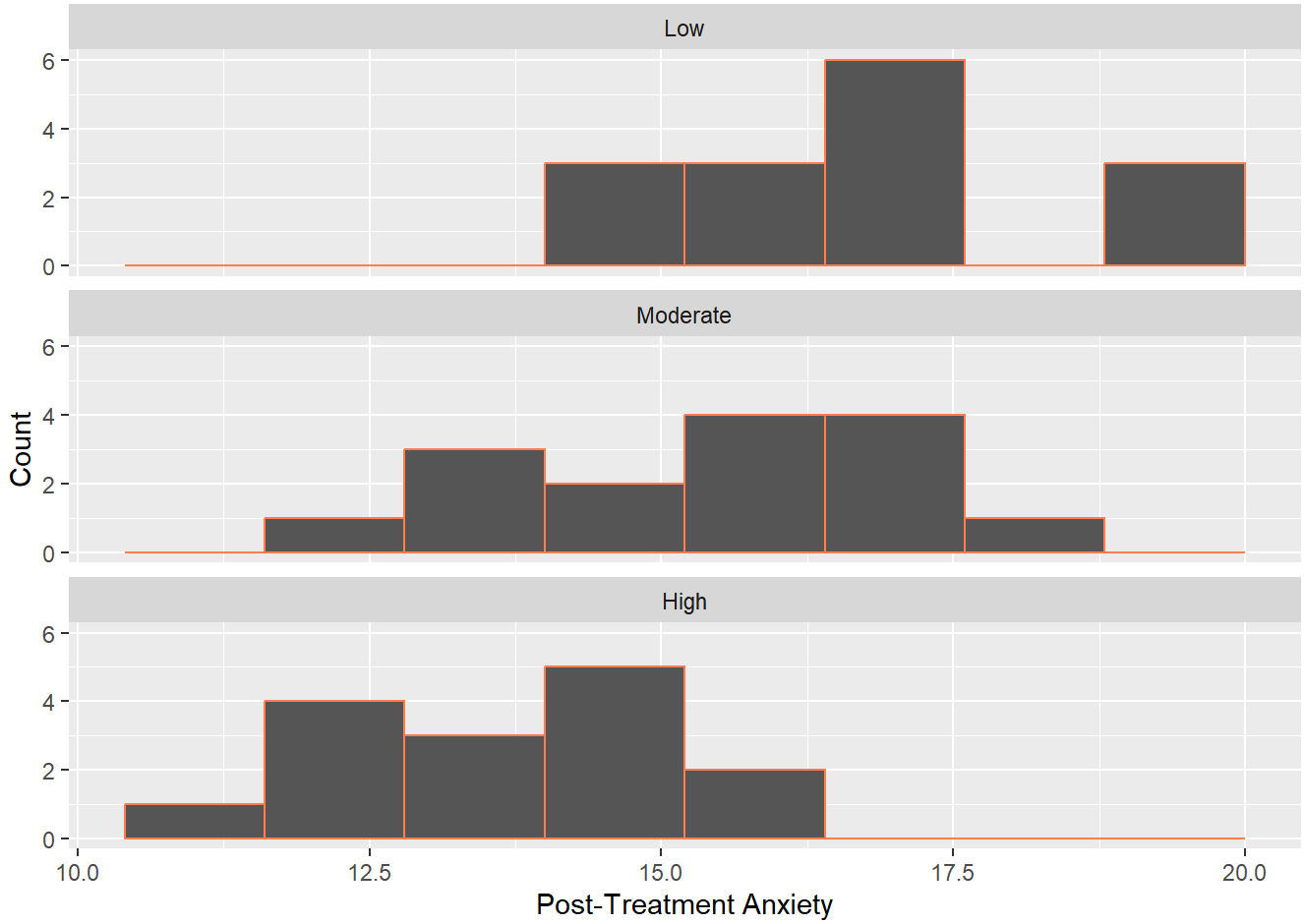
```
##     item   group1 vars  n     mean       sd median  trimmed     mad  min  max
## X11    1      Low    1 15 16.62000 1.675538   16.5 16.60000 1.48260 14.1 19.4
## X12    2 Moderate    1 15 15.52667 1.702715   16.1 15.57692 1.77912 12.7 17.7
## X13    3     High    1 15 13.56000 1.422674   13.8 13.60769 1.92738 11.0 15.5
##     range       skew  kurtosis        se
## X11   5.3  0.2618597 -1.087983 0.4326221
## X12   5.0 -0.3236715 -1.539690 0.4396391
## X13   4.5 -0.3246574 -1.338458 0.3673327
```

ANCOVA | EDA Visualiations

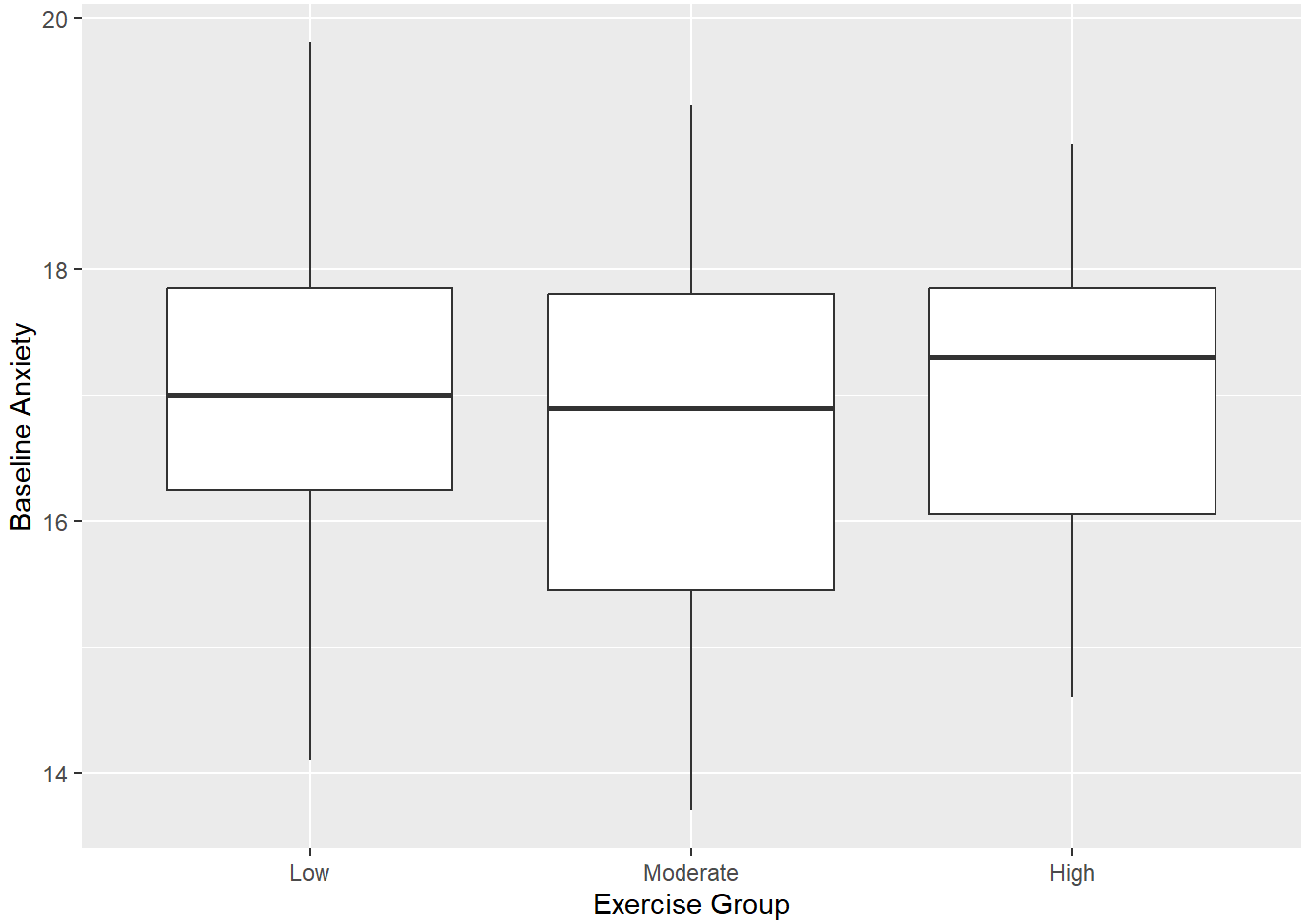
```
# Histograms
ggplot(data = dat_anx,
       mapping = aes(x = pretest)) +
  geom_histogram(bins = 8, color = "coral") +
  labs(y = "Count", x = "Baseline Anxiety")+
  facet_wrap(dat_anx$groupF,ncol=1)
```



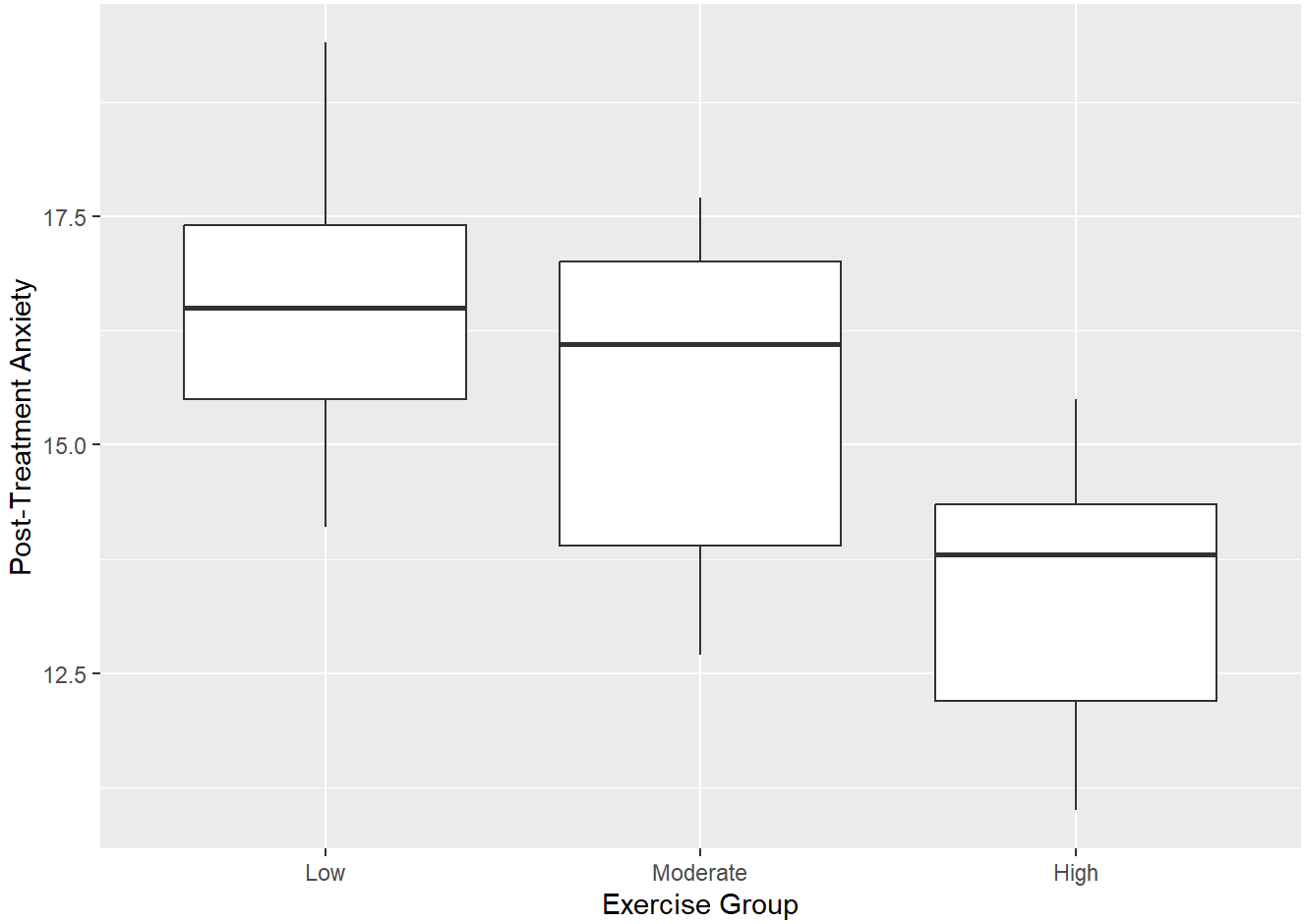

```
ggplot(data = dat_anx,
       mapping = aes(x = posttest)) +
  geom_histogram(bins = 8, color = "coral") +
  labs(y = "Count", x = "Post-Treatment Anxiety")+
  facet_wrap(dat_anx$groupF,ncol=1)
```



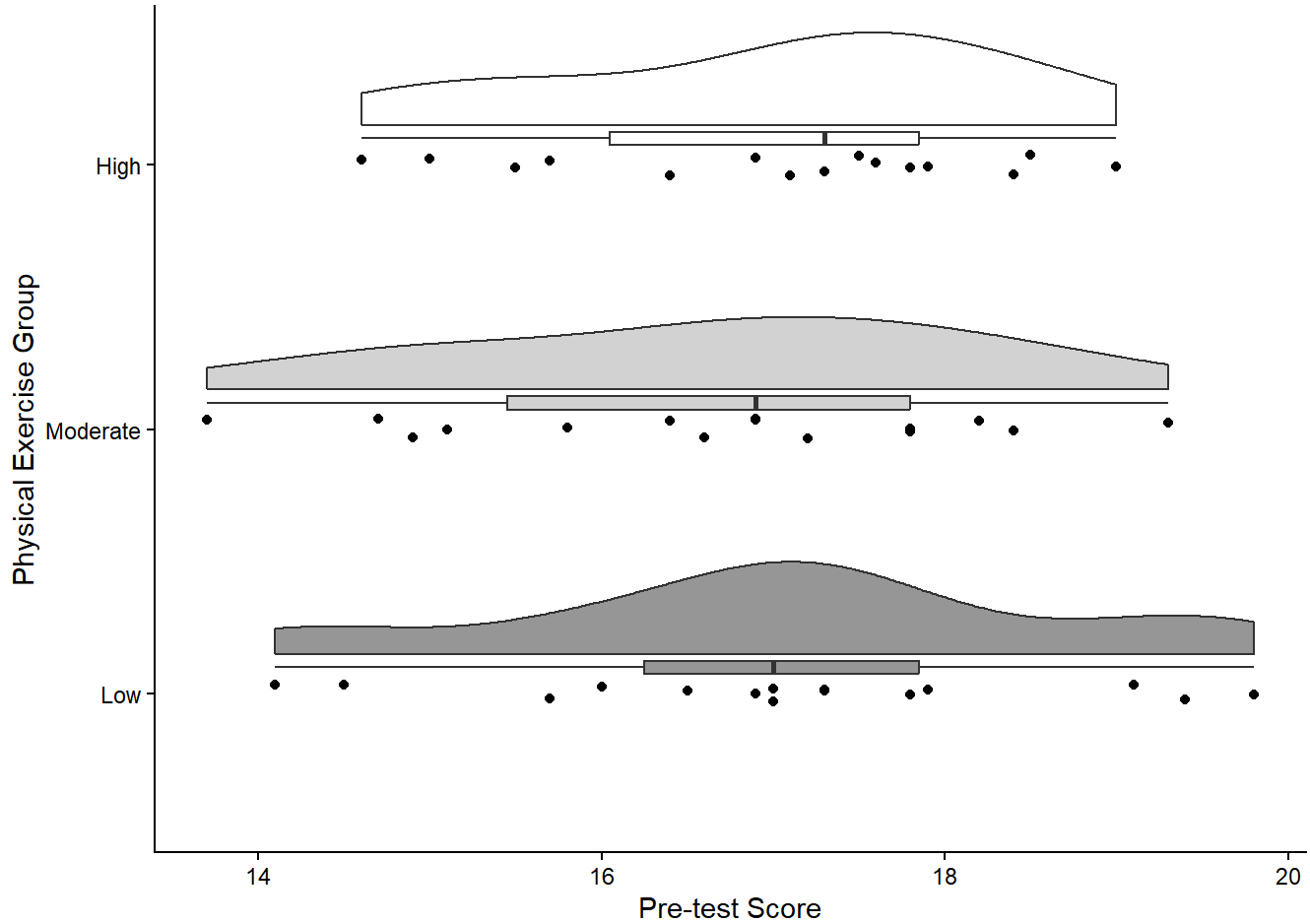

```
# Boxplots
ggplot(data = dat_anx,
       mapping = aes(y = pretest, x = groupF)) +
  geom_boxplot() +
    labs(y = "Baseline Anxiety",
         x = "Exercise Group")
```



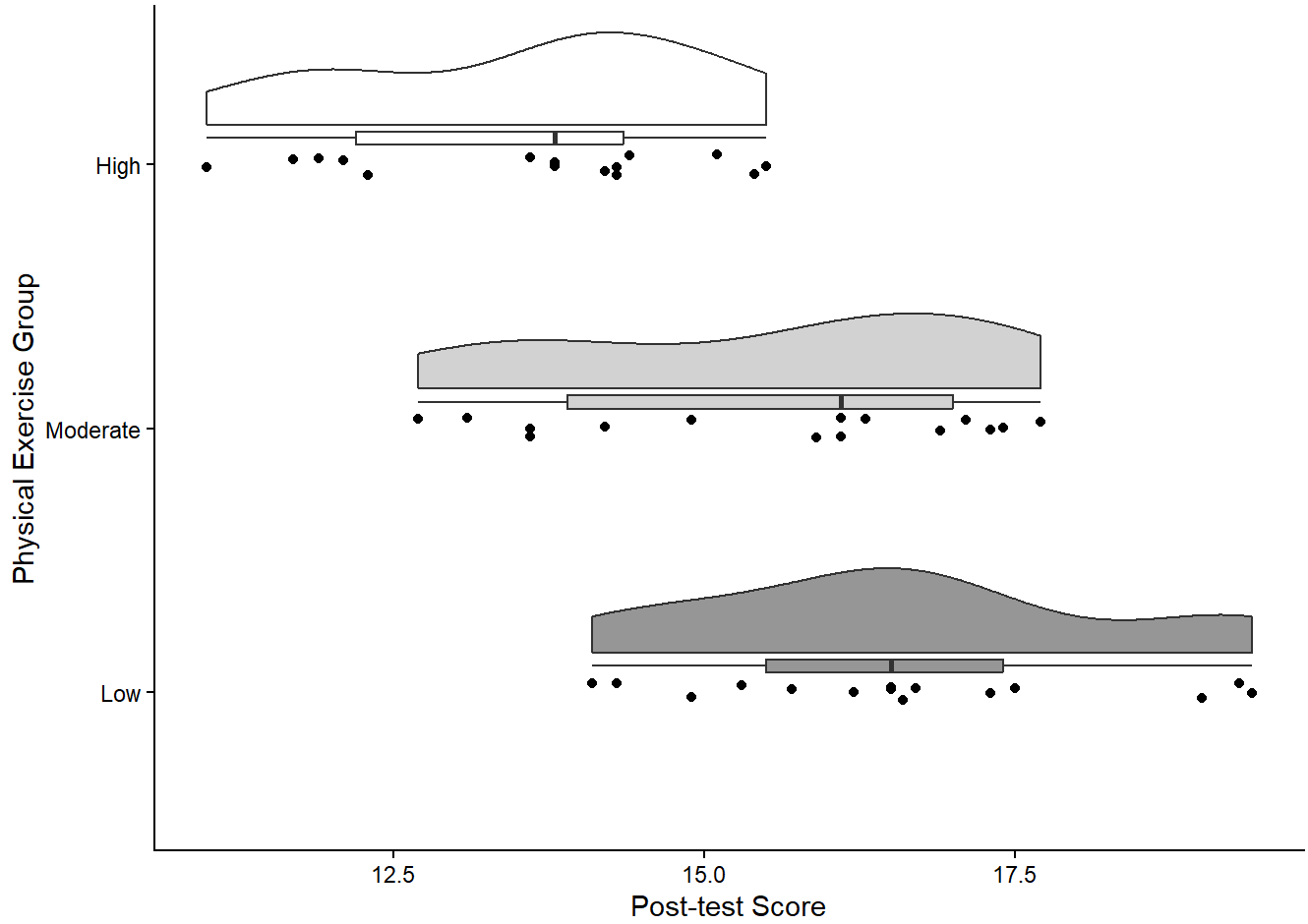

```
ggplot(data = dat_anx,
       mapping = aes(y = posttest, x = groupF)) +
  geom_boxplot() +
    labs(y = "Post-Treatment Anxiety",
         x = "Exercise Group")
```




```
# Raincloud Plots

ggplot(data = dat_anx,
       aes(x = groupF, y = pretest,
           fill = groupF))+
  geom_rain(show.legend=FALSE) +
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Pre-test Score")+
  xlab("Physical Exercise Group")+
  theme_classic()
```




```
ggplot(data = dat_anx,
       aes(x = groupF, y = posttest,
           fill = groupF))+
  geom_rain(show.legend=FALSE) +
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Post-test Score")+
  xlab("Physical Exercise Group")+
  theme_classic()
```

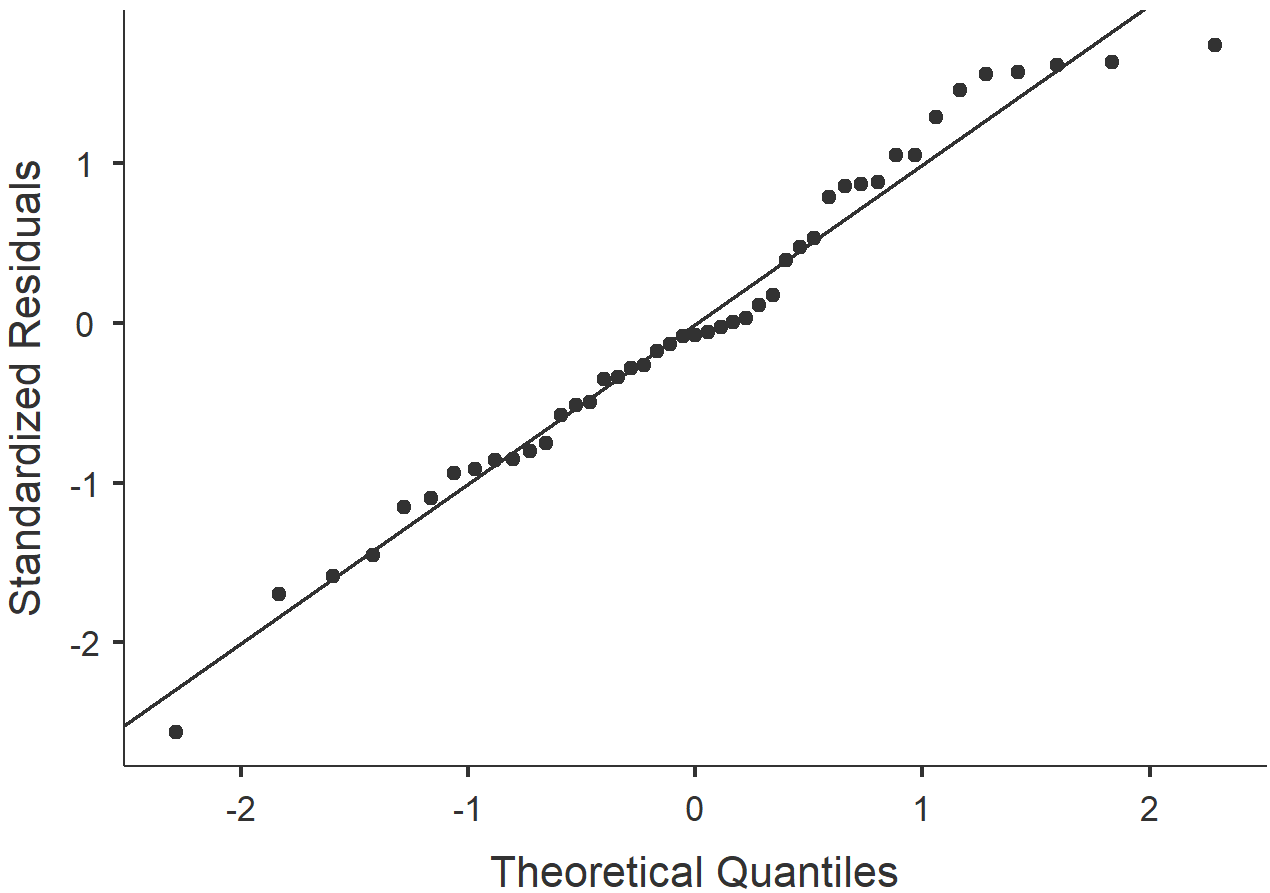


ANCOVA | Omnibus Assumption Check

```
ancova(data = dat_anx,
       dep = posttest,
       factors = groupF,
       covs = pretest,
       effectSize = "eta",
       homo = TRUE,
       qq = TRUE,
       norm = TRUE)
```

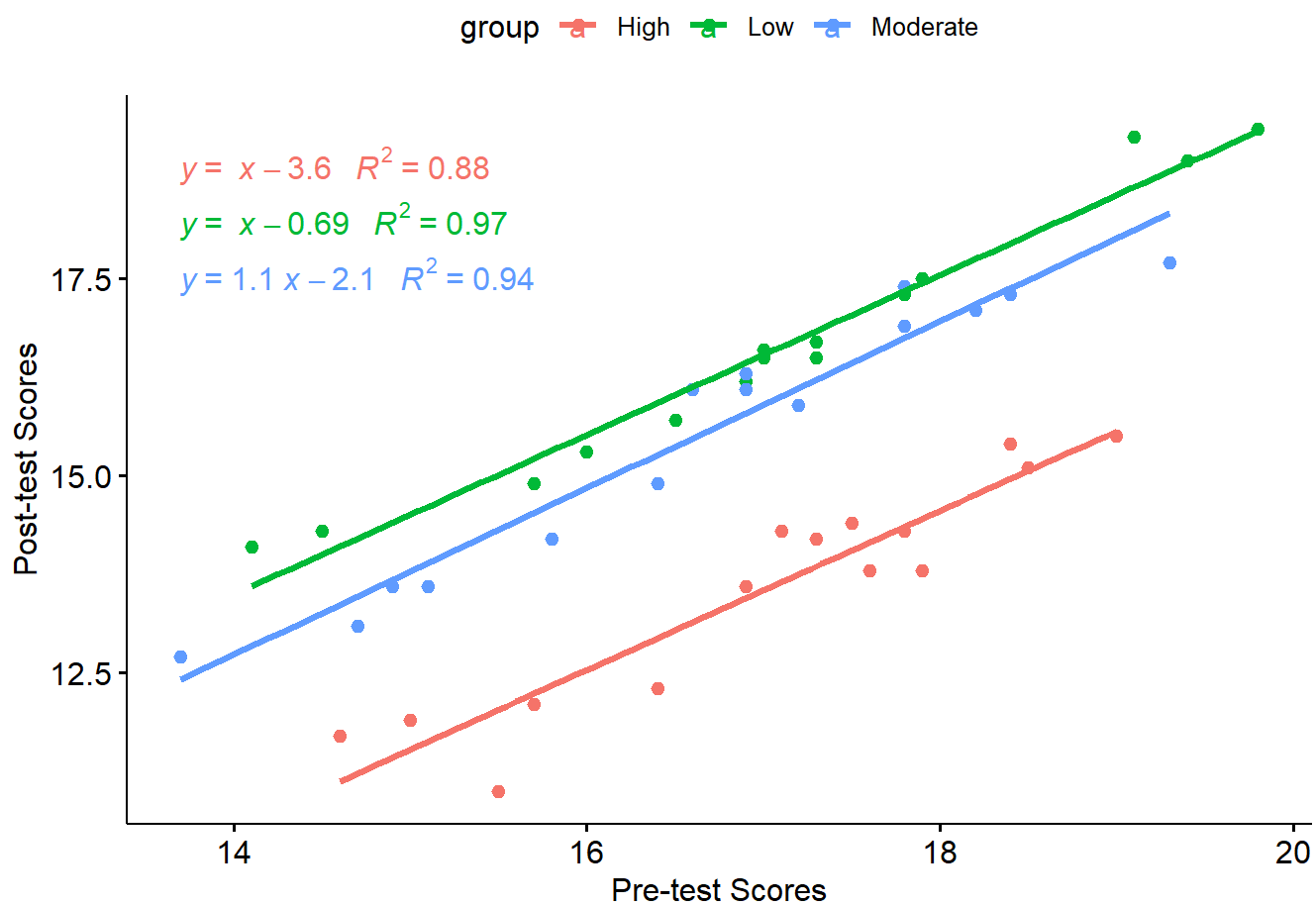
```
##
##  ANCOVA
##
##  ANCOVA - posttest
##  ─────────────────────────────────────────────────────────────────────────────────────────
##                  Sum of Squares    df    Mean Square    F           p             η²
##  ─────────────────────────────────────────────────────────────────────────────────────────
##    groupF            74.022594     2     37.0112970    218.6293    < .0000001    0.4061553
##    pretest          101.288528     1    101.2885278    598.3210    < .0000001    0.5557611
##    Residuals          6.940805    41      0.1692879
##  ─────────────────────────────────────────────────────────────────────────────────────────
##
##
##  ASSUMPTION CHECKS
##
##  Homogeneity of Variances Test (Levene's)
##  ────────────────────────────────────────────
##    F           df1    df2    p
##  ────────────────────────────────────────────
##    2.339315      2     42    0.1088316
##  ────────────────────────────────────────────
##
##
##  Normality Test (Shapiro-Wilk)
##  ──────────────────────────────
##    Statistic    p
##  ──────────────────────────────
##    0.9753224    0.4440646
##  ──────────────────────────────
```

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the jmvcore package.
##   Please report the issue at <https://github.com/jamovi/jmvcore/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



ANCOVA | Linearity Assumption Check

```
ggscatter(dat_anx,
          x = "pretest",
          y = "posttest",
          color = "group",
          add = "reg.line",
          xlab = "Pre-test Scores",
          ylab = "Post-test Scores")+
  stat_regline_equation(
    aes(label = paste(after_stat(eq.label),
                      after_stat(rr.label),
                      sep = "~~~"),
        color = group))
```



ANCOVA | Homogeneity of Regression Slopes Assumption Check

```
dat_anx %>%
  anova_test(posttest ~ groupF*pretest)
```

```
## ANOVA Table (type II tests)
##
##            Effect DFn DFd       F        p p<.05   ges
## 1          groupF   2  39 209.314 1.40e-21     * 0.915
## 2         pretest   1  39 572.828 6.36e-25     * 0.936
## 3 groupF:pretest   2  39   0.127 8.81e-01       0.006
```

```
# Since p > .05 for the interaction we can assume homogeneity of regression slopes
```

ANCOVA | Normality of Residuals

```
lm<-lm(posttest~pretest + groupF,
       dat_anx)

model.metrics <- augment(lm)

shapiro_test(model.metrics$.resid)
```

```
## # A tibble: 1 × 3
##   variable             statistic p.value
##   <chr>                    <dbl>   <dbl>
## 1 model.metrics$.resid     0.975   0.444
```

```
# Lack of statistical significant implies that normality of residuals has been achieved
```

ANCOVA | Homogeneity of Variances

```
model.metrics %>%
  levene_test(.resid ~ groupF)
```

```
## # A tibble: 1 × 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     2    42      2.27 0.116
```

```
# Lack of statistical significance here implies that residual variances are homogeneous across b
etween-subject groups
```

ANCOVA | Conducting the ANCOVA

```
options(digits = 3)
ancova(data = dat_anx,
       dep = posttest,
       factors = groupF,
       covs = pretest,
       effectSize = "eta",
       postHoc = 'groupF',
       postHocCorr = 'bonf',
       postHocES = 'd',
       postHocEsCi = TRUE,
       emMeans=list(
          list('groupF')),
       emmTables=TRUE,
       emmPlots=FALSE)
```

```
##
##  ANCOVA
##
##  ANCOVA - posttest
##  ──────────────────────────────────────────────────────────────────────────
##                 Sum of Squares    df    Mean Square    F      p         η²
##  ──────────────────────────────────────────────────────────────────────────
##    groupF                74.02     2         37.011    219    < .001    0.406
##    pretest              101.29     1        101.289    598    < .001    0.556
##    Residuals              6.94    41          0.169
##  ──────────────────────────────────────────────────────────────────────────
##
##
##  POST HOC TESTS
##
##  Post Hoc Comparisons - groupF
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    groupF           groupF      Mean Difference    SE       df      t        p-bonferroni    Cohen's d    Lower    Upper
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    Low         -    Moderate              0.641    0.151    41.0     4.24    < .001    1.56    0.738     2.38
##                -    High                  2.985    0.150    41.0    19.86    < .001    7.25    5.476     9.03
##    Moderate    -    High                  2.344    0.151    41.0    15.52    < .001    5.70    4.225     7.17
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    Note. Comparisons are based on estimated marginal means
##
##
##  ESTIMATED MARGINAL MEANS
##
##  GROUPF
##
##  Estimated Marginal Means - groupF
##  ──────────────────────────────────────────────
##    groupF      Mean    SE       Lower    Upper
##  ──────────────────────────────────────────────
##    Low         16.4    0.106     16.2     16.7
##    Moderate    15.8    0.107     15.6     16.0
##    High        13.5    0.106     13.2     13.7
##  ──────────────────────────────────────────────
```

ANCOVA | Professional ANCOVA Visualization

```
# res.aov<- dat_anx %>%
#   anova_test(dv = posttest,
#              between= group,
#              covariate = pretest,
#              effect.size = 'ges')

# Generates Marginal Means for Visualization
pwc <- dat_anx %>%
  emmeans_test(posttest ~ group,
               covariate = pretest,
               p.adjust.method = "bonferroni") %>%
  add_xy_position(x = "group")

pwc
```

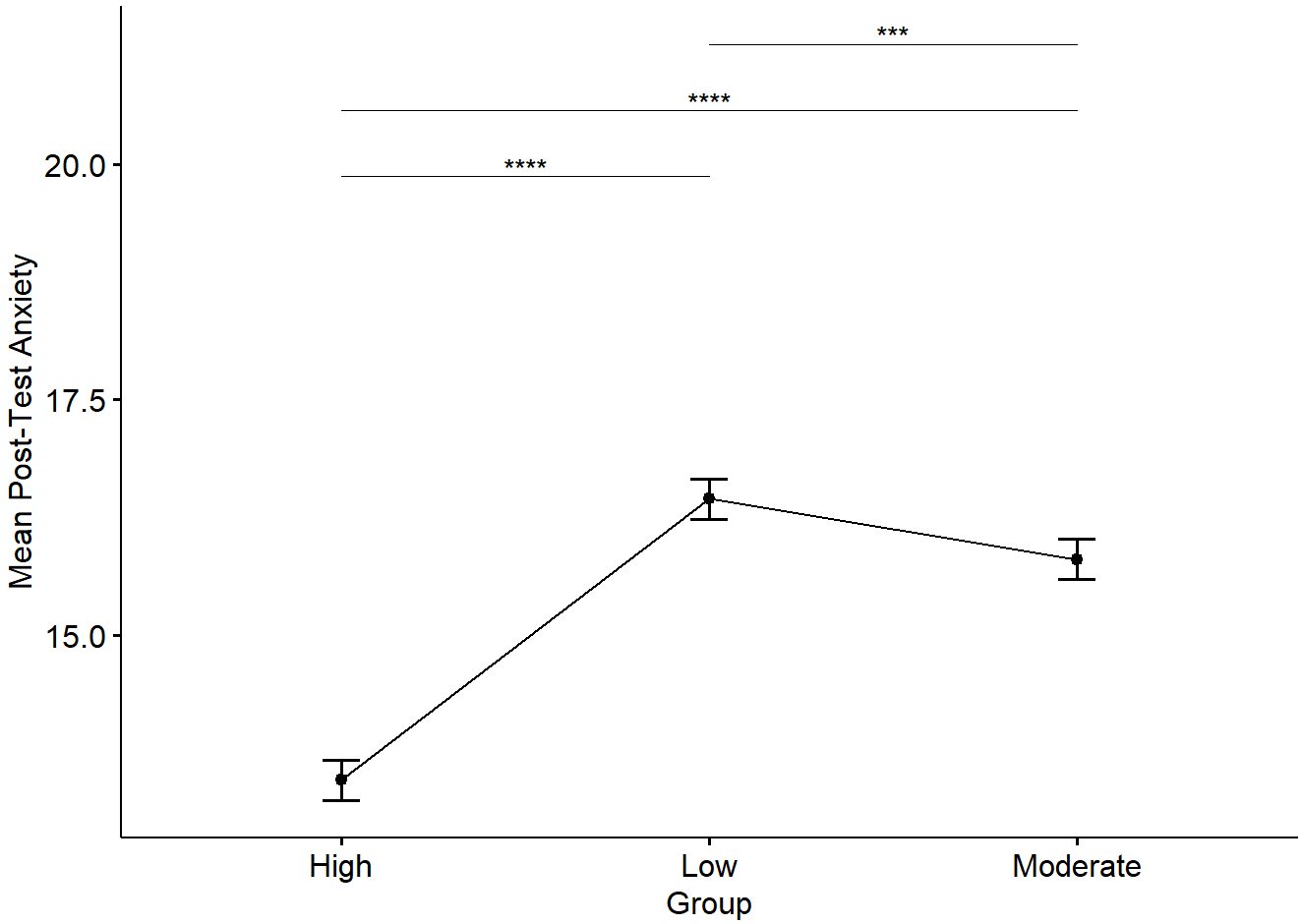
```
## # A tibble: 3 × 13
##   term        .y.   group1 group2    df statistic        p    p.adj p.adj.signif
##   <chr>       <chr> <chr>  <chr>  <dbl>     <dbl>    <dbl>    <dbl> <chr>
## 1 pretest*gr… post… High   Low       41    -19.9  1.19e-22 3.58e-22 ****
## 2 pretest*gr… post… High   Moder…    41    -15.5  9.21e-19 2.76e-18 ****
## 3 pretest*gr… post… Low    Moder…    41      4.24 1.26e- 4 3.77e- 4 ***
## # ℹ 4 more variables: y.position <dbl>, groups <named list>, xmin <dbl>,
## #   xmax <dbl>
```

ANCOVA | Professional ANCOVA Visualization

```
get_emmeans(pwc)
```

```
## # A tibble: 3 × 8
##   pretest group    emmean    se    df conf.low conf.high method
##     <dbl> <fct>     <dbl> <dbl> <dbl>    <dbl>     <dbl> <chr>
## 1    16.9 High       13.5 0.106    41     13.2      13.7 Emmeans test
## 2    16.9 Low        16.4 0.106    41     16.2      16.7 Emmeans test
## 3    16.9 Moderate   15.8 0.107    41     15.6      16.0 Emmeans test
```

```
ggline(get_emmeans(pwc),
       x = "group",
       y = "emmean") +
  geom_errorbar(aes(ymin = conf.low,
                    ymax = conf.high),
                width = 0.1) +
  stat_pvalue_manual(pwc,
                     hide.ns = TRUE,
                     tip.length = FALSE) +
  ylab("Mean Post-Test Anxiety") +
  xlab("Group")
```



ANCOVA | Factorial Research Context Prompt

Your local sports psychologist and stress research enthusiast found out you were learning various quantitative statistical methods and reached out to offer you a consulting/analytics opportunity. They gathered data from 60 participants to attempt to evaluate the effect of a new protein treatment and exercise on experienced stress levels in a group of older adults. The data set (`dat_stress`) contains five variables:

- `id` — Participant identification number
- `score` — Stress score collected from each participant
- `treatment` — Treatment variable with two levels (`yes` — Received protein treatment; `no` — Received placebo treatment)
- `exercise` — Level of exercise employed by participants (`low` , `moderate` , `high`)

- `age` — Age of participant when data was collected

The goal for this final research question was: Does adjusting for the age of participants impact how stress scores fluctuate across treatment and exercise variables?

Factorial ANCOVA | Data

```
dat_stress <- read.csv("ancova2.csv")
```

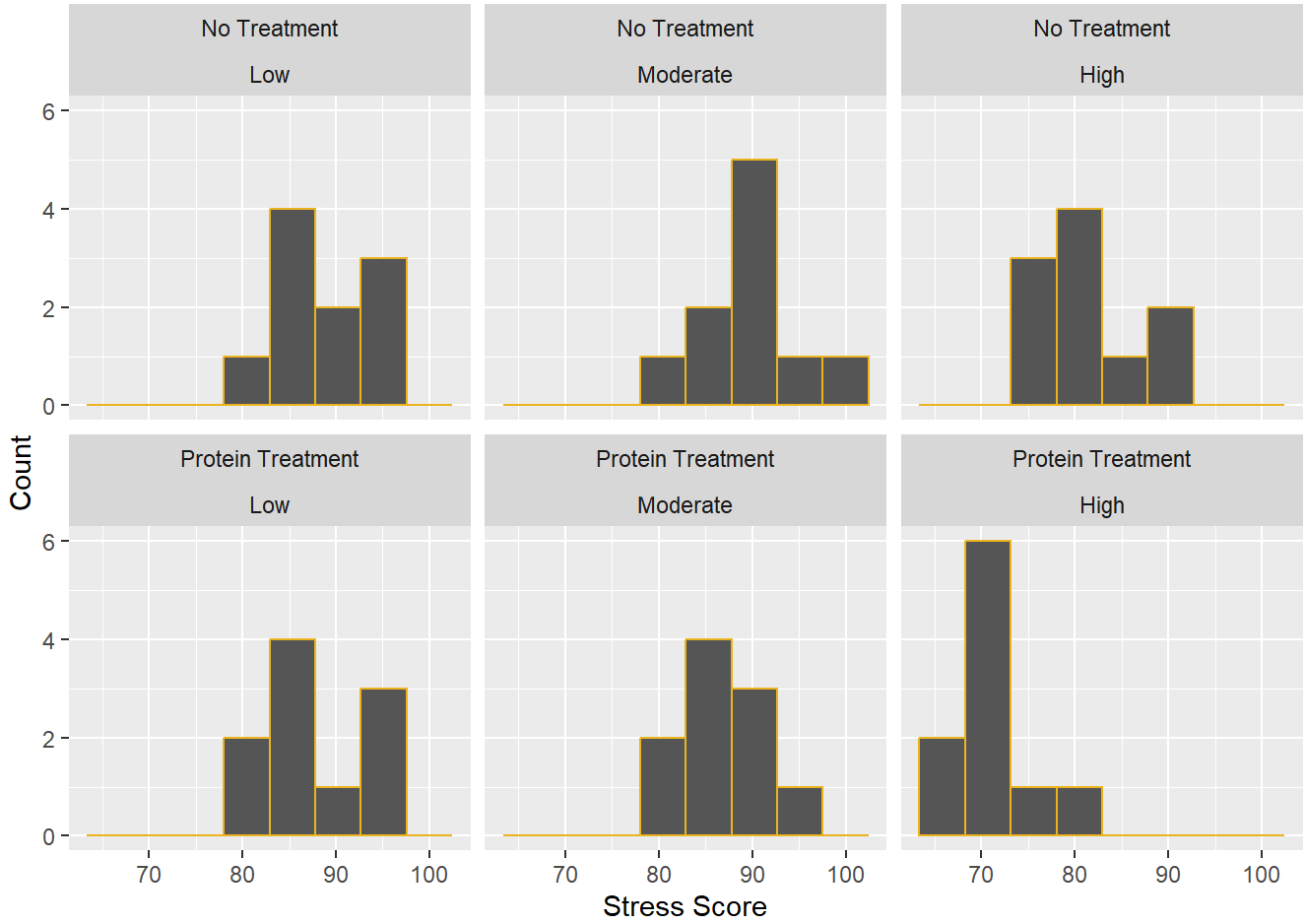
Factorial ANCOVA | Descriptive Stats (Baseline-test)

```
# Descriptive Stats
dat_stress$Treatment <- factor(dat_stress$treatment,
                               levels = c("no", "yes"),
                               labels = c("No Treatment", "Protein Treatment"))

dat_stress$Exercise <- factor(dat_stress$exercise,
                              levels = c("low", "moderate", "high"),
                              labels = c("Low", "Moderate", "High"))

describeBy(dat_stress$score,
           list(dat_stress$Treatment, dat_stress$Exercise),
           mat=TRUE)
```

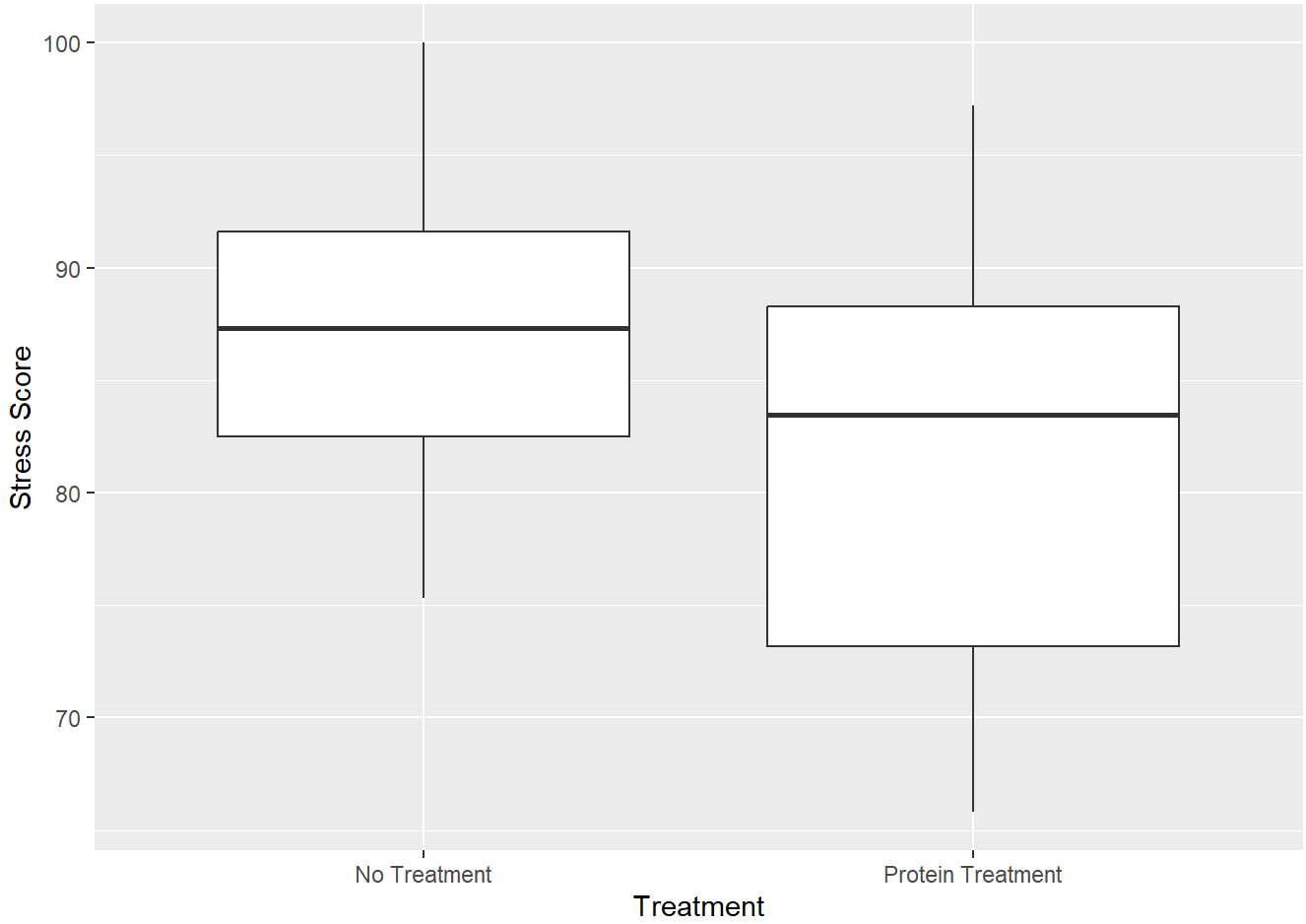
```
##     item            group1   group2 vars  n mean   sd median trimmed  mad  min
## X11    1      No Treatment      Low    1 10 89.6 4.49   89.2    89.7 5.49 82.5
## X12    2 Protein Treatment      Low    1 10 87.9 6.29   84.8    87.5 4.37 81.4
## X13    3      No Treatment Moderate    1 10 89.4 5.42   89.8    89.2 4.23 80.5
## X14    4 Protein Treatment Moderate    1 10 86.8 5.69   87.1    86.6 4.23 78.2
## X15    5      No Treatment     High    1 10 82.0 5.58   82.1    81.6 7.56 75.3
## X16    6 Protein Treatment     High    1 10 71.8 4.33   71.2    71.3 1.93 65.8
##       max range    skew kurtosis   se
## X11  96.1  13.6 -0.0335   -1.565 1.42
## X12  97.2  15.8  0.4837   -1.702 1.99
## X13 100.0  19.5  0.2159   -0.670 1.72
## X14  97.2  19.0  0.0618   -0.916 1.80
## X15  91.7  16.4  0.2737   -1.345 1.77
## X16  81.8  16.0  0.9351    0.303 1.37
```

Factorial ANCOVA | EDA Visualizations

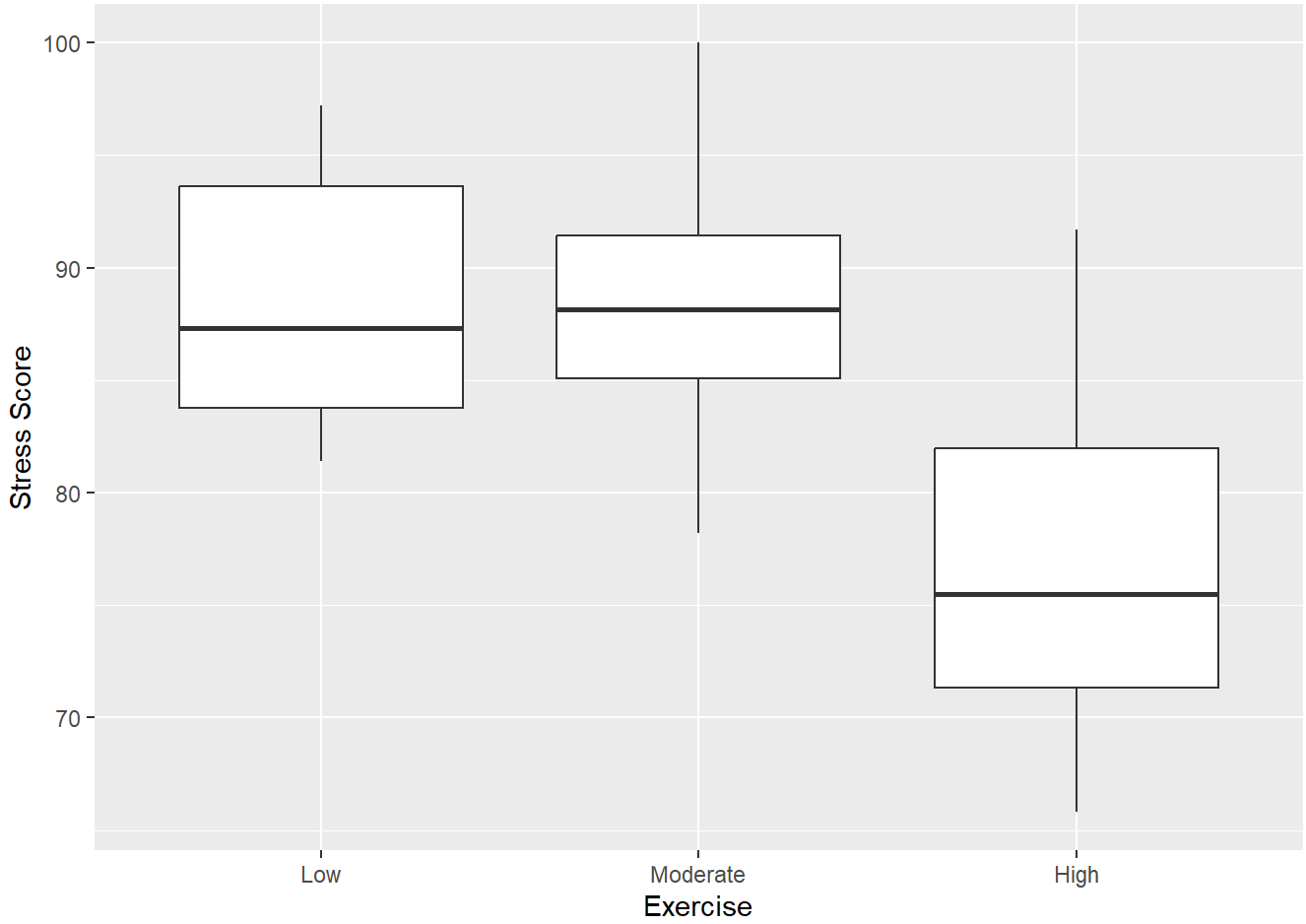
```
# Histogram
ggplot(data = dat_stress,
       mapping = aes(x = score)) +
  geom_histogram(bins = 8, color = "goldenrod2") +
  labs(y = "Count", x = "Stress Score")+
  facet_wrap(~dat_stress$Treatment + dat_stress$Exercise)
```



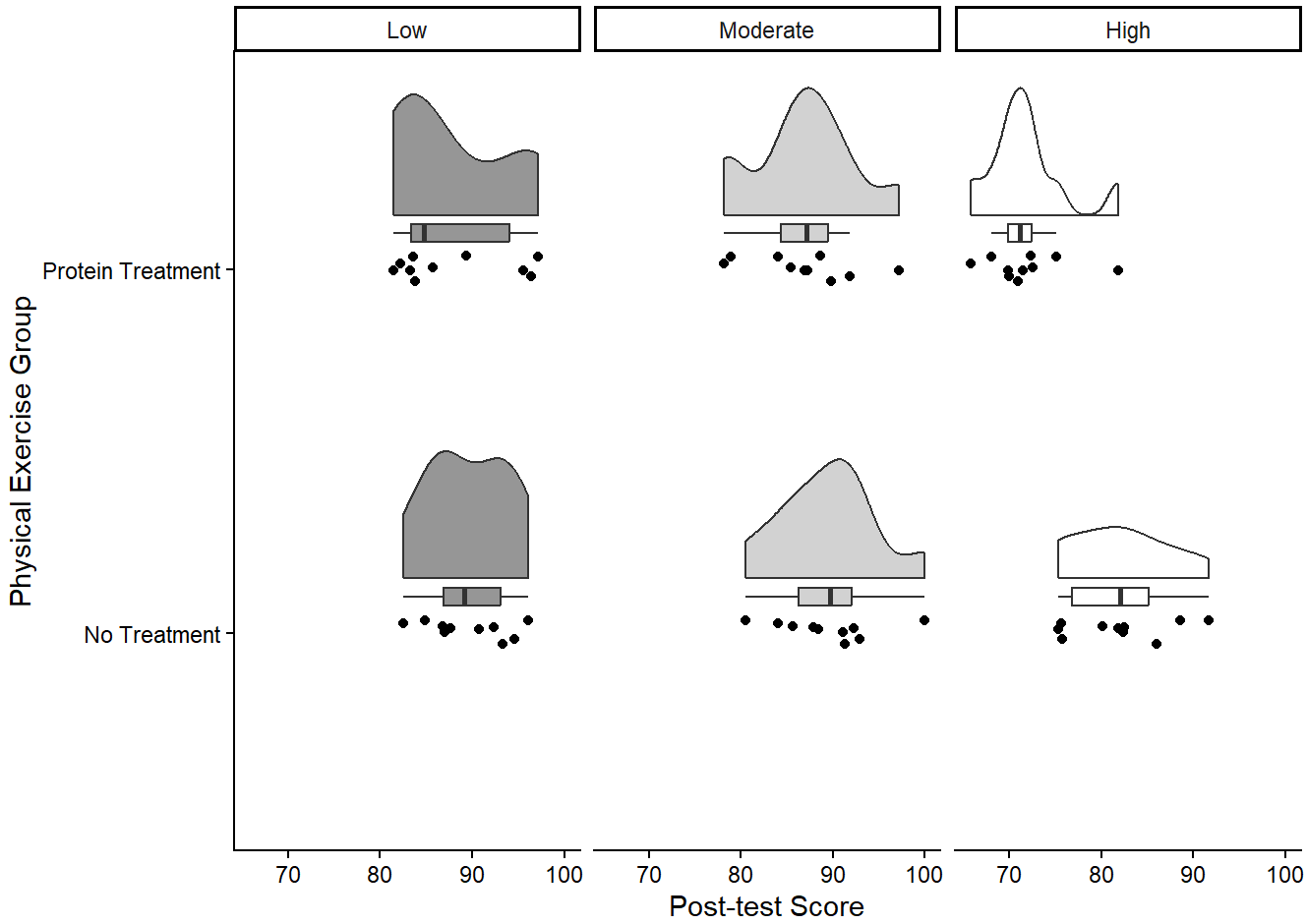

```
# Boxplots
ggplot(data = dat_stress,
       mapping = aes(y = score, x = Treatment)) +
  geom_boxplot() +
    labs(y = "Stress Score")
```



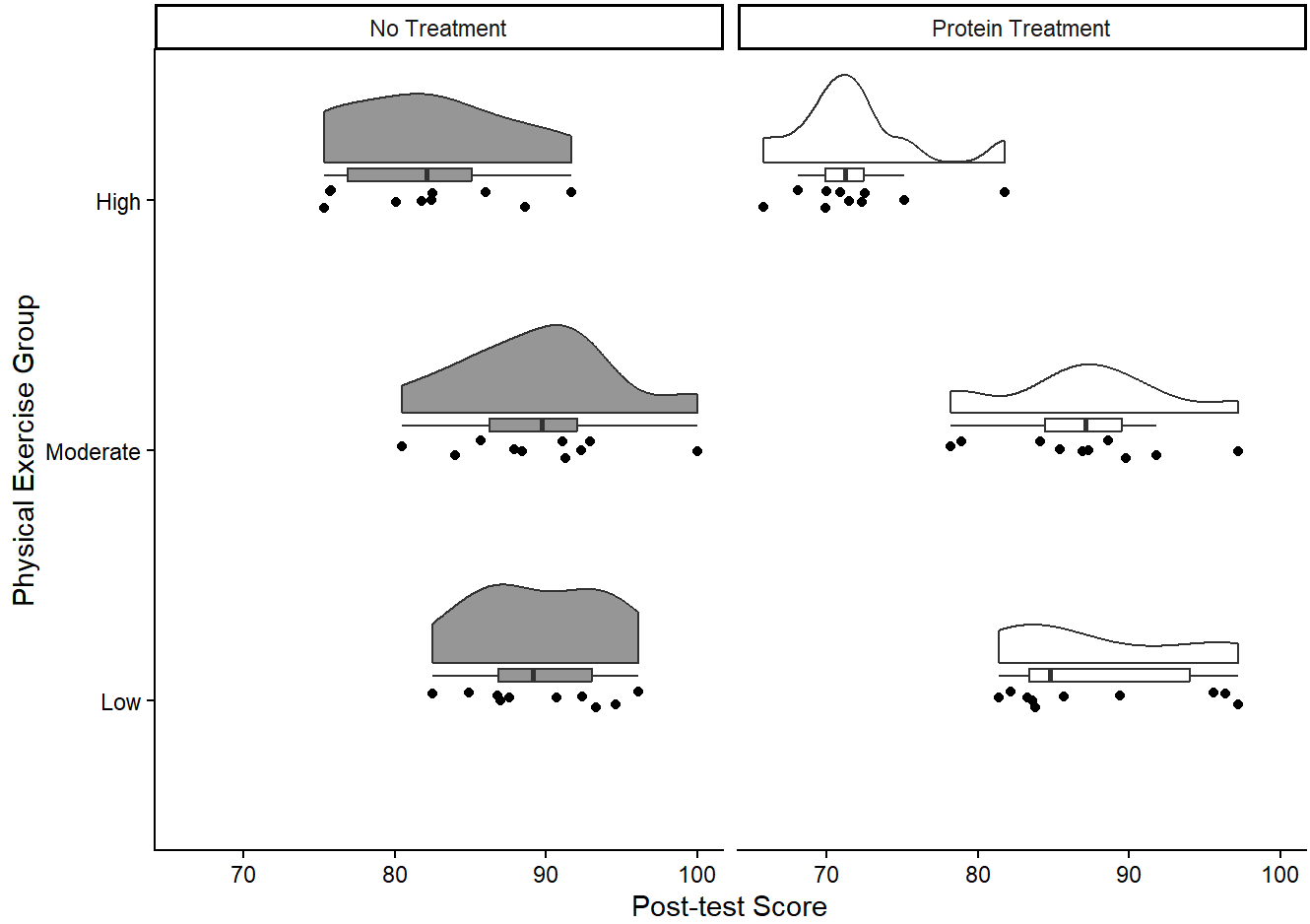

```
ggplot(data = dat_stress,
       mapping = aes(y = score, x = Exercise)) +
  geom_boxplot() +
    labs(y = "Stress Score")
```




```
# Violin Plot

ggplot(data = dat_stress,
       aes(x = Treatment, y = score,
           fill = Exercise))+
  geom_rain(show.legend=FALSE) +
  facet_wrap(~Exercise)+
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Post-test Score")+
  xlab("Physical Exercise Group")+
  theme_classic()
```



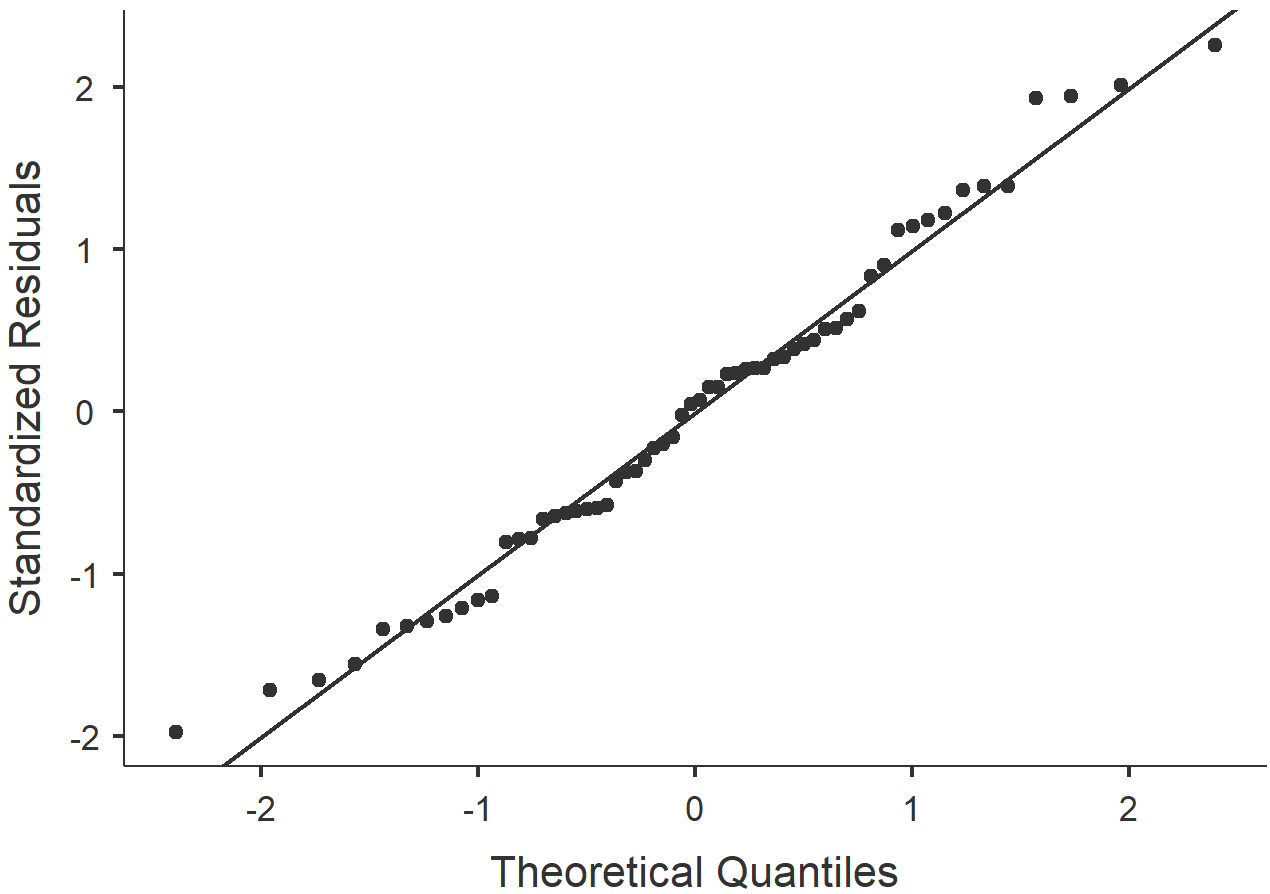

```
ggplot(data = dat_stress,
       aes(x = Exercise, y = score,
           fill = Treatment))+
  geom_rain(show.legend=FALSE) +
  facet_wrap(~Treatment)+
  scale_colour_grey(aesthetics = "fill",
                    start = .6,
                    end = 1) +
  coord_flip() +
  ylab("Post-test Score")+
  xlab("Physical Exercise Group")+
  theme_classic()
```



Factorial ANCOVA | Assumption Check: Omnibus and Grouped Level

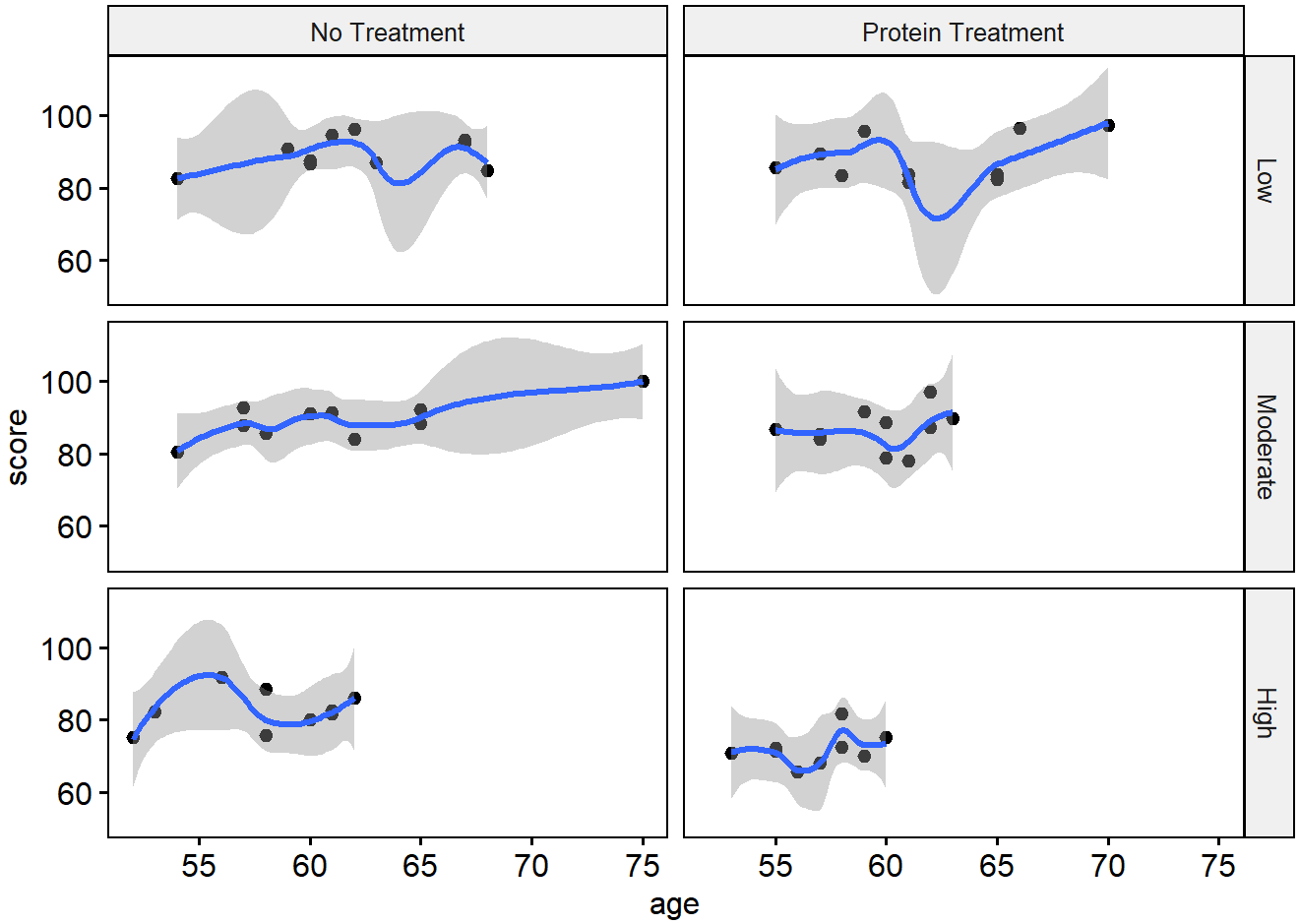
```
# Check Assumptions using techniques described in class

ancova(data = dat_stress,
       dep = score,
       factors = c(Treatment,Exercise),
       covs = age,
       effectSize = "partEta",
       homo = TRUE,
       qq = TRUE,
       norm = TRUE)
```

```
##
##  ANCOVA
##
##  ANCOVA - score
##  ──────────────────────────────────────────────────────────────────────────────────────
##                          Sum of Squares    df    Mean Square    F        p         η²p
##  ──────────────────────────────────────────────────────────────────────────────────────
##    Treatment                        275     1          275.0    11.07     0.002    0.173
##    Exercise                        1029     2          514.7    20.71    < .001    0.439
##    age                              226     1          226.4     9.11     0.004    0.147
##    Treatment:Exercise               221     2          110.5     4.45     0.016    0.144
##    Residuals                       1317    53           24.8
##  ──────────────────────────────────────────────────────────────────────────────────────
##
##
##  ASSUMPTION CHECKS
##
##  Homogeneity of Variances Test (Levene's)
##  ────────────────────────────────────────
##    F        df1    df2    p
##  ────────────────────────────────────────
##    0.955      5     54    0.453
##  ────────────────────────────────────────
##
##
##  Normality Test (Shapiro-Wilk)
##  ─────────────────────────────
##    Statistic    p
##  ─────────────────────────────
##        0.982    0.531
##  ─────────────────────────────
```




```
# Grouped Assumption Checks

# Linear relationship based on age and IVs
ggscatter(dat_stress,
          x = "age",
          y = "score",
          facet.by = c("Exercise","Treatment")) +
  stat_smooth(method = "loess")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Homogeneity of Regression slopes

dat_stress %>%
  anova_test(score ~ age + Treatment + Exercise + Treatment*Exercise + age*Treatment + age*Exerc
ise + age*Exercise*Treatment)
```

```
## ANOVA Table (type II tests)
##
##                    Effect DFn DFd      F        p p<.05      ges
## 1                     age   1  48  8.359 6.00e-03     * 0.148000
## 2               Treatment   1  48  9.907 3.00e-03     * 0.171000
## 3                Exercise   2  48 18.197 1.31e-06     * 0.431000
## 4     Treatment:Exercise   2  48  3.303 4.50e-02     * 0.121000
## 5           age:Treatment   1  48  0.009 9.25e-01       0.000189
## 6            age:Exercise   2  48  0.235 7.91e-01       0.010000
## 7 age:Treatment:Exercise   2  48  0.073 9.30e-01       0.003000
```

```
# Normality assessments
lm<-lm(score~age + Exercise*Treatment, dat_stress)
model.metrics <- augment(lm)
shapiro_test(model.metrics$.resid)
```

```
## # A tibble: 1 × 3
##   variable             statistic p.value
##   <chr>                    <dbl>   <dbl>
## 1 model.metrics$.resid     0.982   0.531
```

```
# Homogeneity of variance assessments
model.metrics %>%
  levene_test(.resid ~ Exercise*Treatment)
```

```
## # A tibble: 1 × 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     5    54     0.829 0.535
```

Factorial ANCOVA | Conducting the ANCOVA

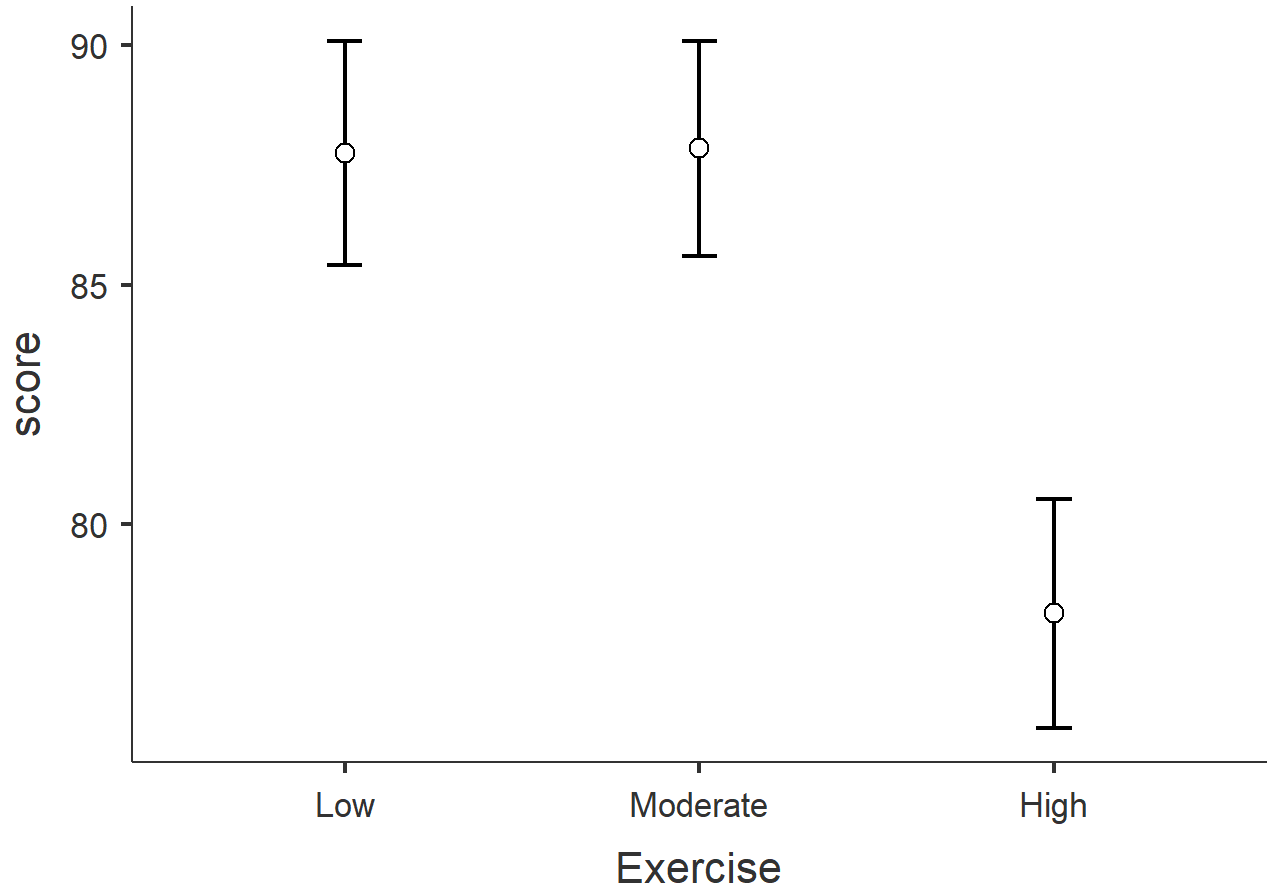
```
options(digits=3)
ancova(data = dat_stress,
       dep = score,
       factors = c(Treatment,Exercise),
       covs = age,
       effectSize = "partEta",
       postHoc = ~Exercise + Treatment:Exercise,
       postHocCorr = 'bonf',
       postHocES = 'd',
       postHocEsCi = TRUE,
       emMeans= ~ Exercise + Treatment + Exercise:Treatment,
       emmTables=TRUE,
       emmPlots=TRUE)
```

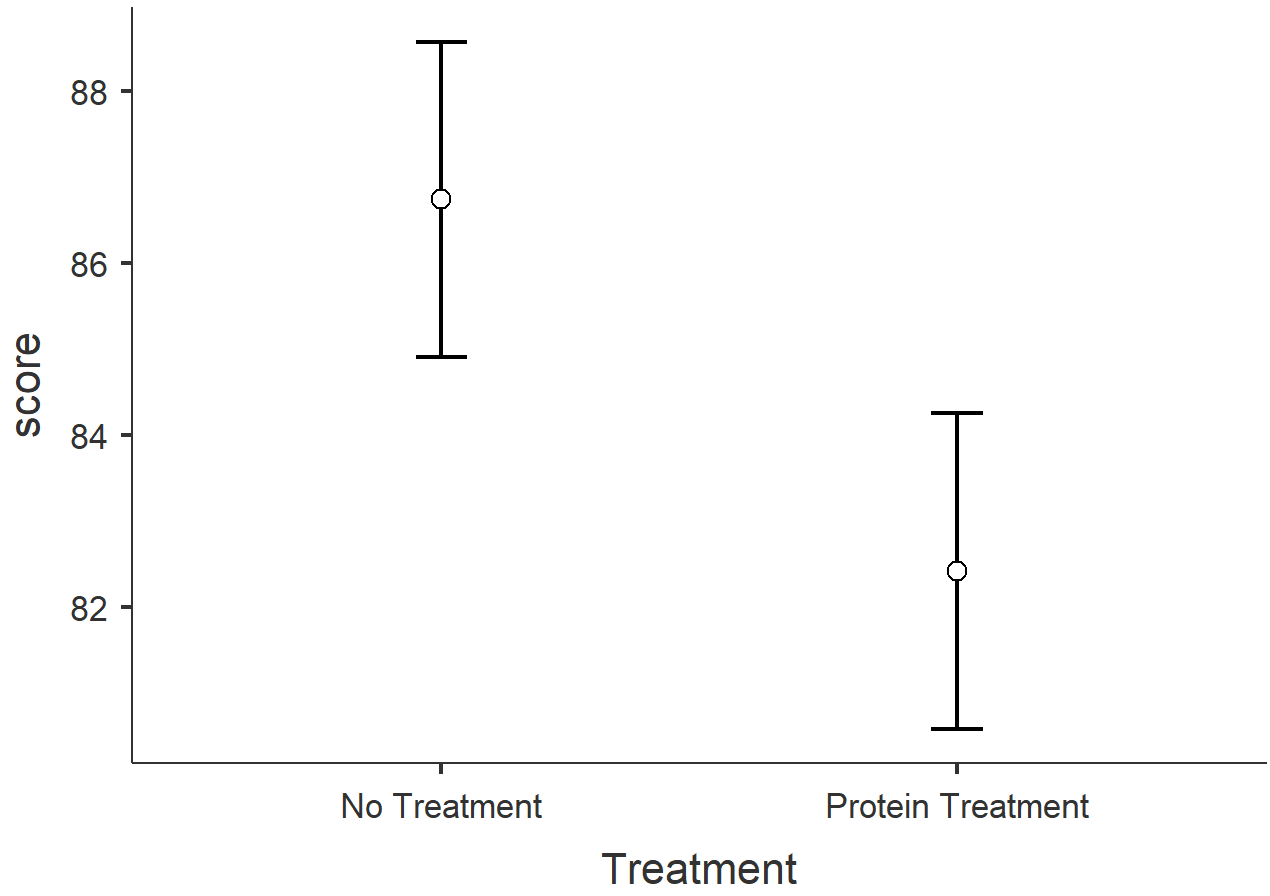
```
## NOTE: Results may be misleading due to involvement in interactions
```

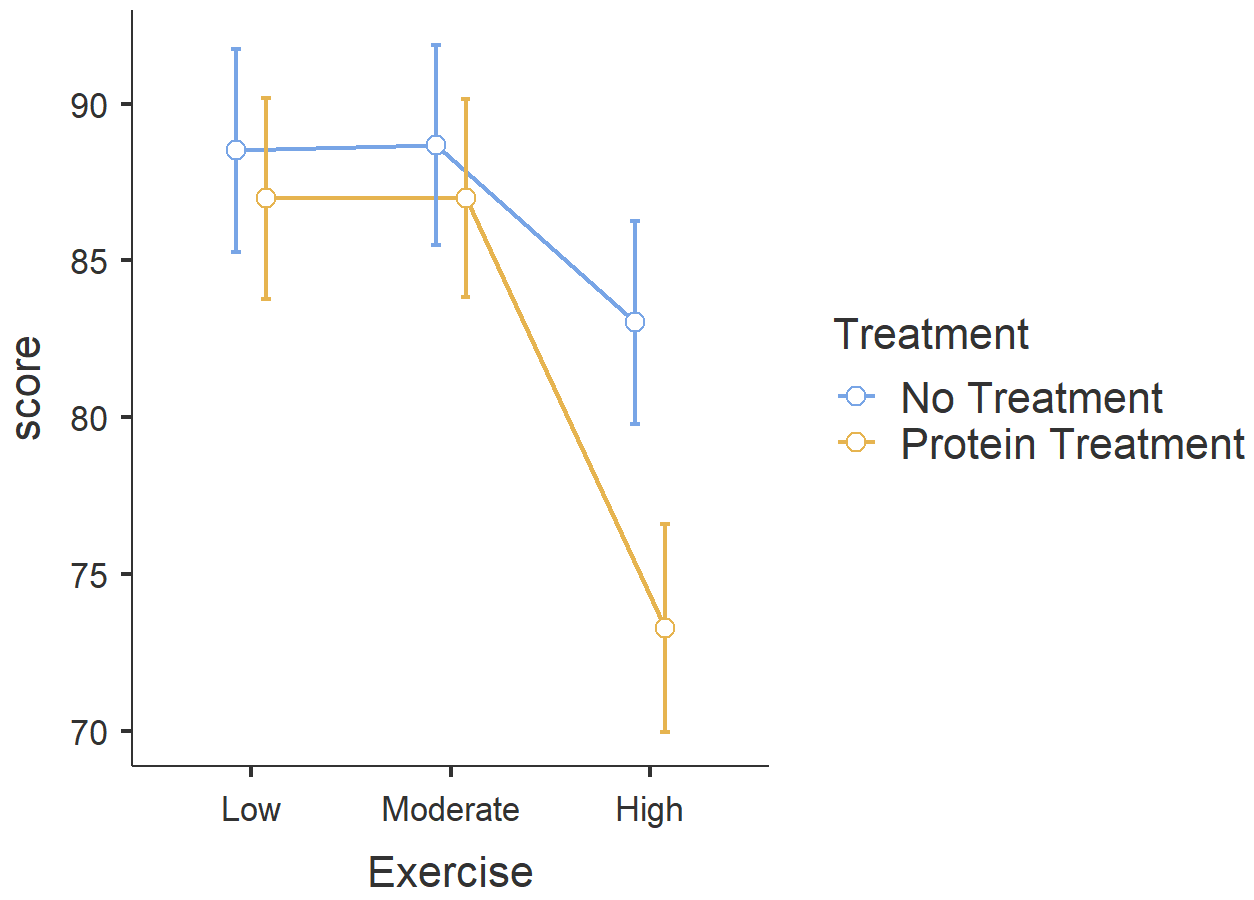
```
##
##  ANCOVA
##
##  ANCOVA - score
##  ──────────────────────────────────────────────────────────────────────────────────────────
##                           Sum of Squares    df    Mean Square    F        p         η²p
##  ──────────────────────────────────────────────────────────────────────────────────────────
##    Treatment                        275     1          275.0    11.07     0.002    0.173
##    Exercise                        1029     2          514.7    20.71    < .001    0.439
##    age                              226     1          226.4     9.11     0.004    0.147
##    Treatment:Exercise               221     2          110.5     4.45     0.016    0.144
##    Residuals                       1317    53           24.8
##  ──────────────────────────────────────────────────────────────────────────────────────────
##
##
##  POST HOC TESTS
##
##  Post Hoc Comparisons - Exercise
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    Exercise         Exercise    Mean Difference    SE      df      t          p-bonferroni
Cohen's d    Lower     Upper
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    Low         -    Moderate            -0.0950    1.59    53.0    -0.0596           1.000
-0.0191    -0.660    0.622
##                -    High                 9.5942    1.74    53.0     5.5063    < .001       1.
9247     1.130    2.720
##    Moderate    -    High                 9.6892    1.66    53.0     5.8495    < .001       1.
9438     1.177    2.710
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
##    Note. Comparisons are based on estimated marginal means
##
##
##  Post Hoc Comparisons - Treatment:Exercise
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────
##    Treatment            Exercise         Treatment            Exercise    Mean Difference
SE      df      t           p-bonferroni    Cohen's d    Lower     Upper
##  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────
##    No Treatment         Low         -    No Treatment         Moderate            -0.1725
2.23    53.0    -0.07727           1.000     -0.03460    -0.933    0.864
##                                     -    No Treatment         High                 5.4851
2.34    53.0     2.34727           0.340      1.10037     0.136    2.065
##                                     -    Protein Treatment    Low                  1.5286
2.23    53.0     0.68538           1.000      0.30665    -0.593    1.206
##                                     -    Protein Treatment    Moderate             1.5111
2.27    53.0     0.66630           1.000      0.30315    -0.611    1.218
##                                     -    Protein Treatment    High                15.2319
2.39    53.0     6.38354    < .001      3.05569     1.926    4.185
```

```
##                          Moderate    -    No Treatment         High                 5.6576
2.30    53.0     2.45505           0.261      1.13497     0.182    2.088
##                                       -    Protein Treatment    Low                  1.7011
2.23    53.0     0.76287           1.000     -0.34125    -1.241    0.558
##                                       -    Protein Treatment    Moderate             1.6836
2.25    53.0     0.74847           1.000      0.33775    -0.570    1.245
##                                       -    Protein Treatment    High                15.4044
2.35    53.0     6.56340    < .001      3.09029     1.970    4.210
##                          High        -    Protein Treatment    Low                 -3.9565
2.32    53.0    -1.70712           1.000      0.79372    -0.152    1.739
##                                       -    Protein Treatment    Moderate            -3.9740
2.25    53.0    -1.76839           1.000      0.79722    -0.120    1.715
##                                       -    Protein Treatment    High                 9.7468
2.23    53.0     4.36234    < .001      1.95532     0.979    2.932
##    Protein Treatment    Low         -    Protein Treatment    Moderate            -0.0175
2.26    53.0    -0.00774           1.000     -0.00350    -0.912    0.905
##                                       -    Protein Treatment    High                13.7033
2.36    53.0     5.79876    < .001      2.74904     1.658    3.840
##                          Moderate    -    Protein Treatment    High                13.7208
2.27    53.0     6.04155    < .001      2.75254     1.693    3.812
## ────────────
```

```
##  Estimated Marginal Means - Exercise:Treatment
## ──────────────────────────────────────────────────────────────────────
##   Treatment            Exercise    Mean    SE      Lower    Upper
## ──────────────────────────────────────────────────────────────────────
##   No Treatment         Low         88.5    1.62     85.3     91.7
##                        Moderate    88.7    1.59     85.5     91.9
##                        High        83.0    1.61     79.8     86.3
##   Protein Treatment    Low         87.0    1.60     83.8     90.2
##                        Moderate    87.0    1.58     83.8     90.2
##                        High        73.3    1.65     70.0     76.6
## ──────────────────────────────────────────────────────────────────────
```


90
85
80
score
Low
Moderate
High
Exercise


88
86
84
82
score
No Treatment
Protein Treatment
Treatment




```
# Assessment of Simple Main Effects
dat_stress %>%
  group_by(Exercise) %>%
  anova_test(score ~ age + Treatment)
```

```
## # A tibble: 6 × 8
##   Exercise Effect      DFn   DFd      F        p `p<.05`   ges
## * <fct>    <chr>     <dbl> <dbl>  <dbl>    <dbl> <chr>   <dbl>
## 1 Low      age           1    17  2.25  0.152    ""      0.117
## 2 Low      Treatment     1    17  0.437 0.517    ""      0.025
## 3 Moderate age           1    17  6.65  0.02     "*"     0.281
## 4 Moderate Treatment     1    17  0.419 0.526    ""      0.024
## 5 High     age           1    17  0.794 0.385    ""      0.045
## 6 High     Treatment     1    17 18.7   0.000455 "*"     0.524
```

```
dat_stress %>%
  group_by(Treatment) %>%
  anova_test(score ~ age + Exercise)
```

```
## # A tibble: 4 × 8
##   Treatment         Effect     DFn   DFd     F         p `p<.05`   ges
## * <fct>             <chr>    <dbl> <dbl> <dbl>     <dbl> <chr>   <dbl>
## 1 No Treatment      age          1    26  7.26 0.012     "*"     0.218
## 2 No Treatment      Exercise     2    26  3.99 0.031     "*"     0.235
## 3 Protein Treatment age          1    26  2.37 0.136     ""      0.083
## 4 Protein Treatment Exercise     2    26 17.3  0.0000164 "*"     0.572
```

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```
# Pairwise Comparisons
pwc <- dat_stress %>%
  group_by(Exercise) %>%
  emmeans_test(score ~ Treatment,
               covariate = age,
               p.adjust.method = "bonferroni")
pwc
```

```
## # A tibble: 3 × 10
##   Exercise term          .y.   group1     group2    df statistic       p   p.adj
## * <fct>    <chr>         <chr> <chr>      <chr>  <dbl>     <dbl>   <dbl>   <dbl>
## 1 Low      age*Treatment score No Treatm… Prote…    53     0.685 4.96e-1 4.96e-1
## 2 Moderate age*Treatment score No Treatm… Prote…    53     0.748 4.57e-1 4.57e-1
## 3 High     age*Treatment score No Treatm… Prote…    53     4.36  5.97e-5 5.97e-5
## # ℹ 1 more variable: p.adj.signif <chr>
```

```
pwc %>% filter(Exercise == "High")
```

```
## # A tibble: 1 × 10
##   Exercise term          .y.   group1     group2    df statistic       p   p.adj
##   <fct>    <chr>         <chr> <chr>      <chr>  <dbl>     <dbl>   <dbl>   <dbl>
## 1 High     age*Treatment score No Treatm… Prote…    53      4.36 5.97e-5 5.97e-5
## # ℹ 1 more variable: p.adj.signif <chr>
```

```
pwc2 <- dat_stress %>%
  group_by(Treatment) %>%
  emmeans_test(score ~ Exercise,
               covariate = age,
               p.adjust.method = "bonferroni")
pwc2
```

```
## # A tibble: 6 × 10
##   Treatment         term     .y.   group1 group2    df statistic       p   p.adj
## * <fct>             <chr>    <chr> <chr>  <chr>  <dbl>     <dbl>   <dbl>   <dbl>
## 1 No Treatment      age*Exe… score Low    Moder…    53  -0.0773  9.39e-1 1   e+0
## 2 No Treatment      age*Exe… score Low    High      53   2.35    2.27e-2 6.80e-2
## 3 No Treatment      age*Exe… score Moder… High      53   2.46    1.74e-2 5.22e-2
## 4 Protein Treatment age*Exe… score Low    Moder…    53  -0.00774 9.94e-1 1   e+0
## 5 Protein Treatment age*Exe… score Low    High      53   5.80    3.78e-7 1.13e-6
## 6 Protein Treatment age*Exe… score Moder… High      53   6.04    1.55e-7 4.66e-7
## # ℹ 1 more variable: p.adj.signif <chr>
```

```
pwc2 %>% filter(Treatment == "Protein Treatment")
```

```
## # A tibble: 3 × 10
##   Treatment         term     .y.   group1 group2    df statistic       p   p.adj
##   <fct>             <chr>    <chr> <chr>  <chr>  <dbl>     <dbl>   <dbl>   <dbl>
## 1 Protein Treatment age*Exe… score Low    Moder…    53  -0.00774 9.94e-1 1   e+0
## 2 Protein Treatment age*Exe… score Low    High      53   5.80    3.78e-7 1.13e-6
## 3 Protein Treatment age*Exe… score Moder… High      53   6.04    1.55e-7 4.66e-7
## # ℹ 1 more variable: p.adj.signif <chr>
```

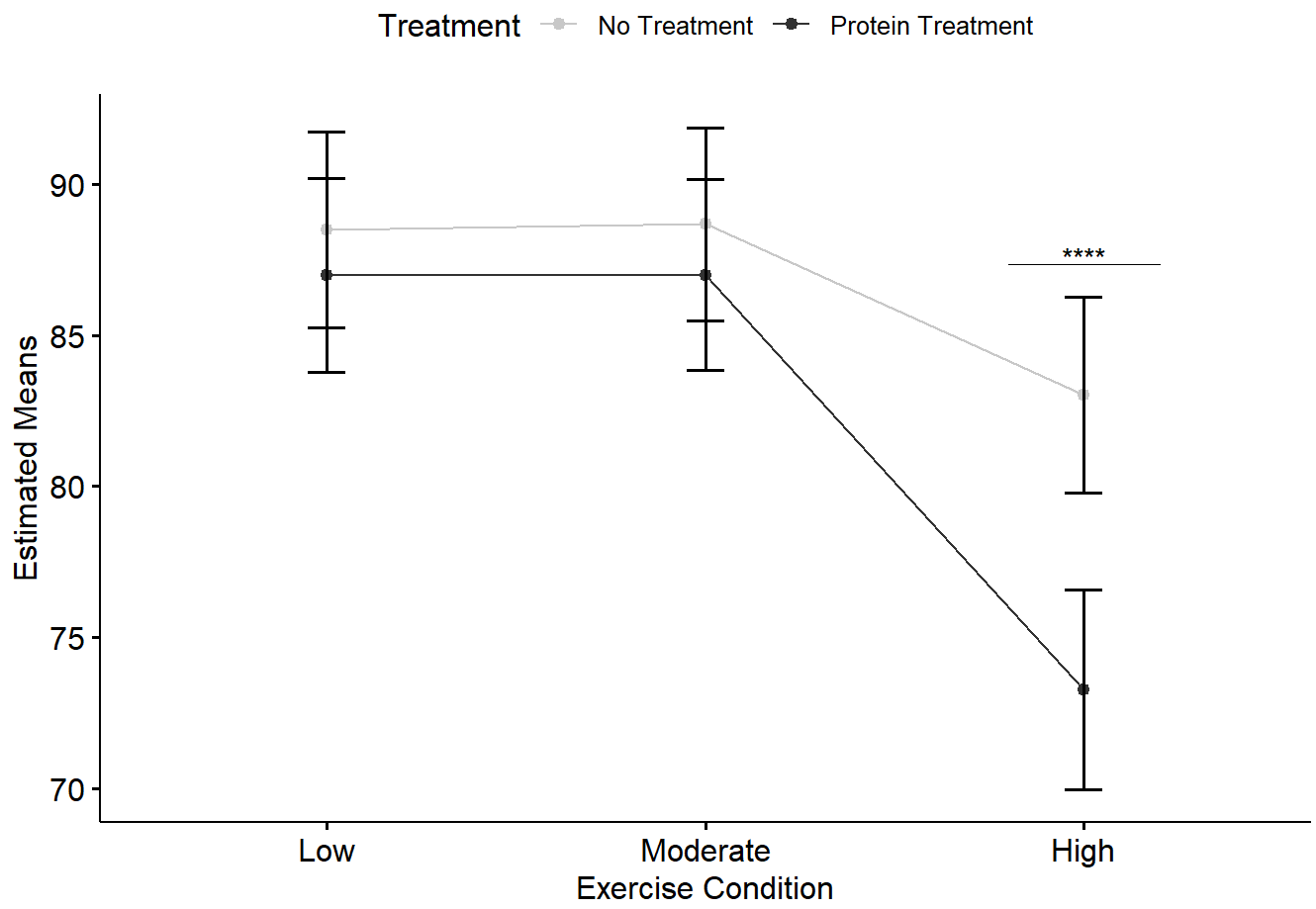
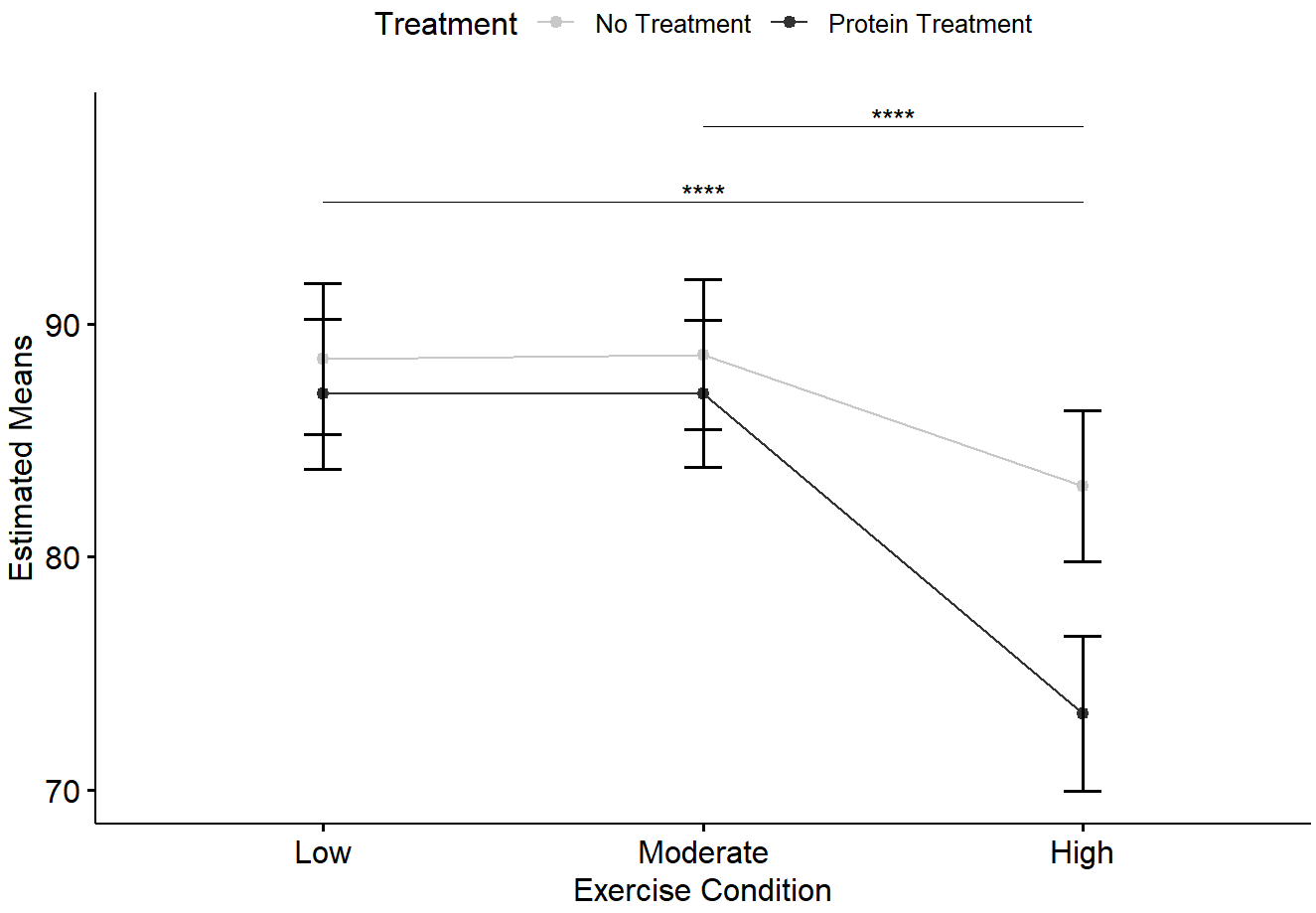
```
lp <- ggline(
  get_emmeans(pwc),
  x = "Exercise",
  y = "emmean",
  color = "Treatment",
  palette = "grey") +
  geom_errorbar(
    aes(ymin = conf.low,
        ymax = conf.high),
    width = 0.1)

lp2 <- ggline(
  get_emmeans(pwc2),
  x = "Exercise",
  y = "emmean",
  color = "Treatment",
  palette = "grey") +
  geom_errorbar(
    aes(ymin = conf.low,
        ymax = conf.high),
    width = 0.1)
```

```
pwc <- pwc %>% add_xy_position(x = "Exercise",
                   fun = "mean_se",
                   step.increase = 0.2)

pwc.filtered <- pwc %>% filter(Exercise == "High")


lp + stat_pvalue_manual(pwc,
                      hide.ns = TRUE,
                      tip.length = 0) +
  labs(
    y = "Estimated Means",
    x = "Exercise Condition",
    color = "Treatment"
  )
```



```
pwc2 <- pwc2 %>% add_xy_position(x = "Exercise",
                  fun = "mean_se")

pwc2.filtered <- pwc2 %>% filter(Treatment == "Protein Treatment")


lp2 + stat_pvalue_manual(pwc2,
                         hide.ns = TRUE,
                         tip.length = 0) +
  labs(
    y = "Estimated Means",
    x = "Exercise Condition",
    color = "Treatment"
  )
```